

World Models

Lab Seminar 260710



강화학습(Reinforcement Learning)에 관하여

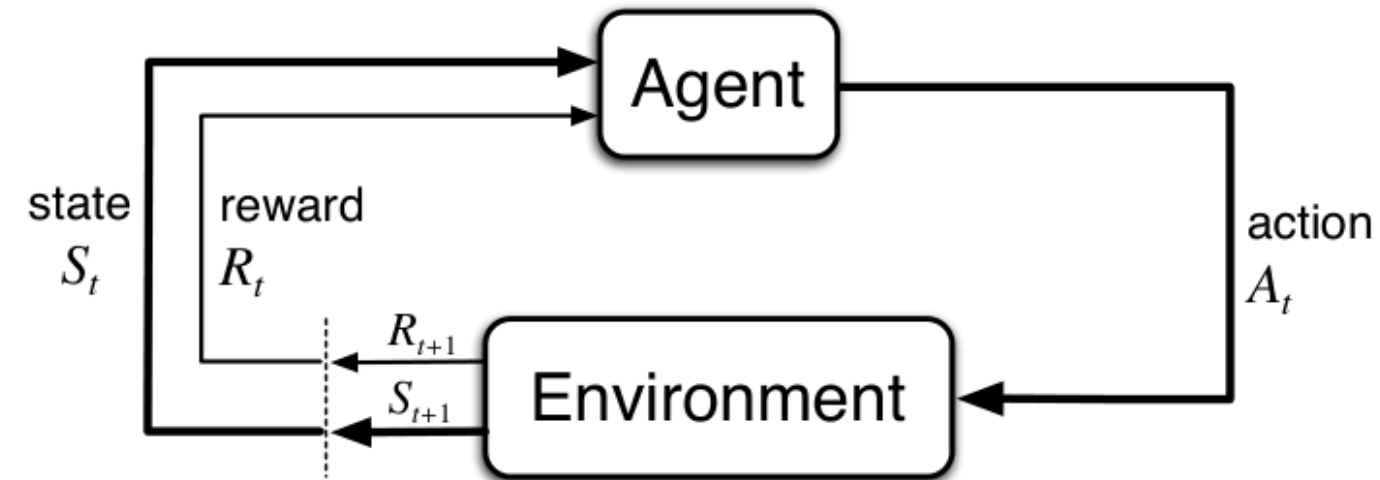


Figure 3.1: The agent–environment interaction in a Markov decision process.

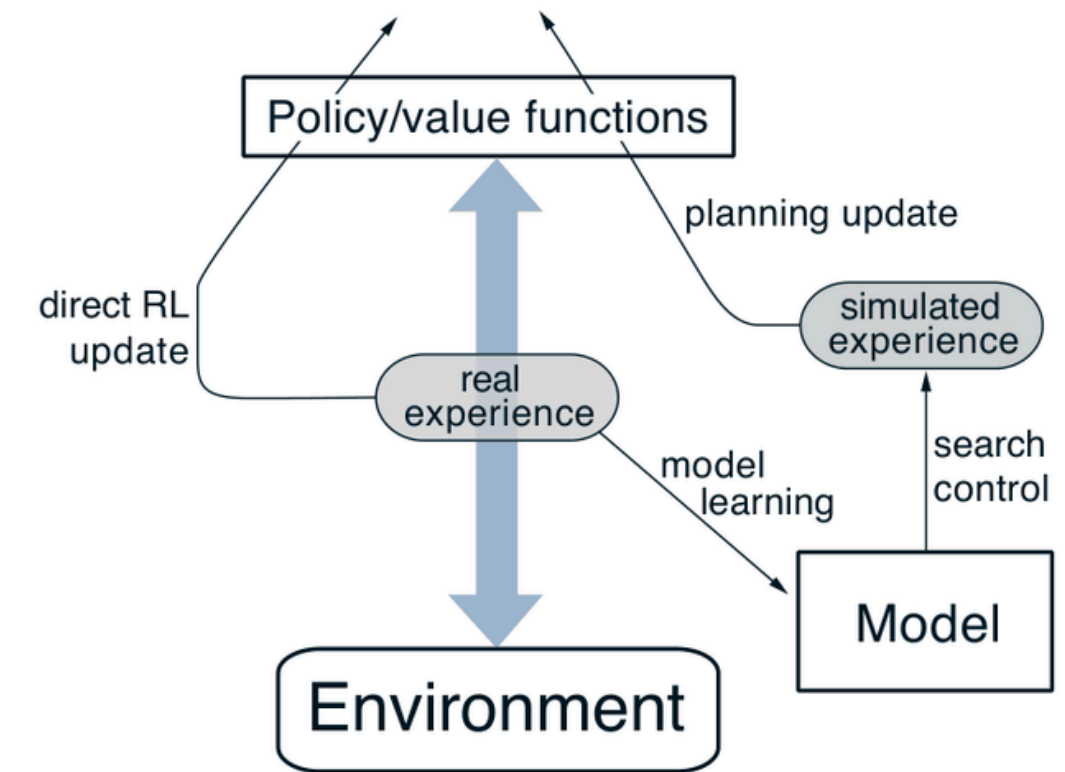


Figure 8.1: The general Dyna Architecture. Real experience, passing back and forth between the environment and the policy, affects policy and value functions in much the same way as does simulated experience generated by the model of the environment.

월드모델이란?

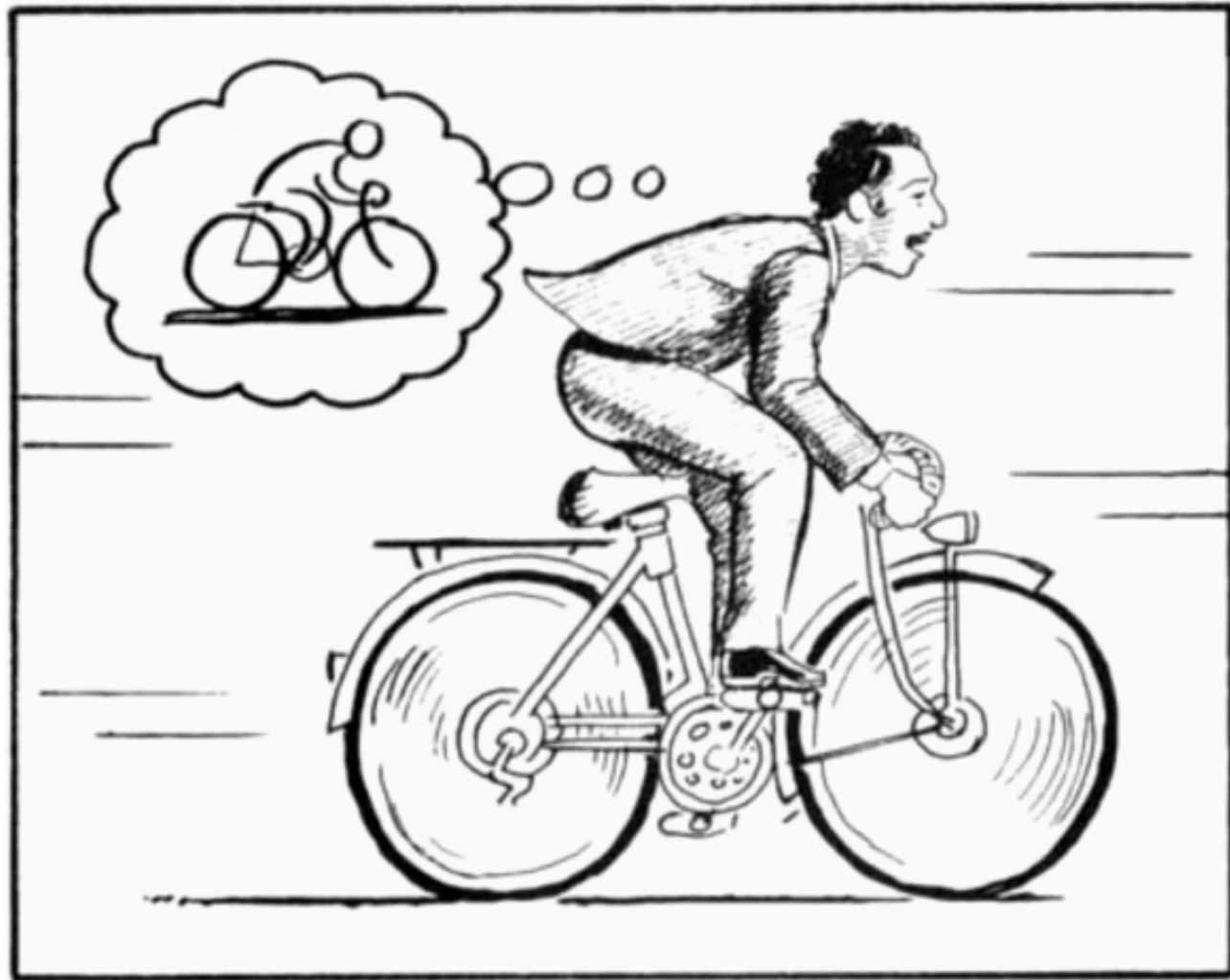
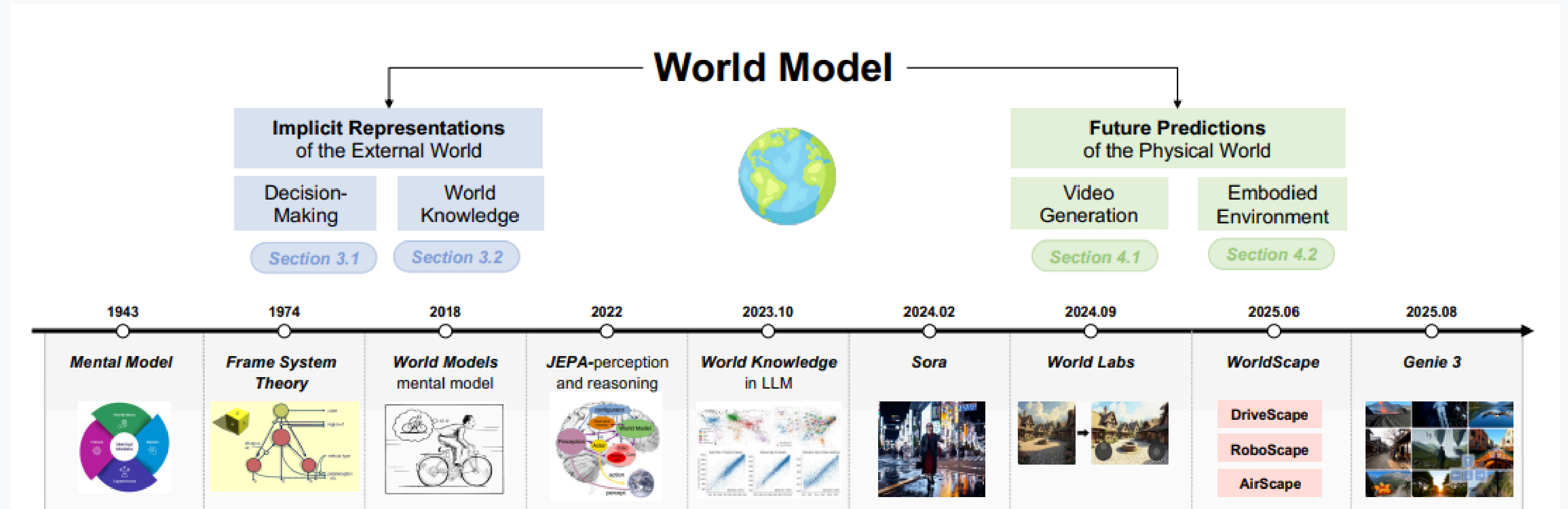


Figure 1. A World Model, from Scott McCloud's *Understanding Comics*. (McCloud, 1993; E, 2012)

현재 환경 + 행동 → 다음 환경

쉽게 말하자면 어떻게 하면 세상을
시뮬레이션 할 수 있을까?

월드모델이란?



World Models

World Models

David Ha¹ Jürgen Schmidhuber^{2,3}

Abstract

We explore building generative neural network models of popular reinforcement learning environments. Our *world model* can be trained quickly in an unsupervised manner to learn a compressed spatial and temporal representation of the environment. By using features extracted from the world model as inputs to an agent, we can train a very compact and simple policy that can solve the required task. We can even train our agent entirely inside of its own hallucinated dream generated by its world model, and transfer this policy back into the actual environment.

An interactive version of this paper is available at
<https://worldmodels.github.io>

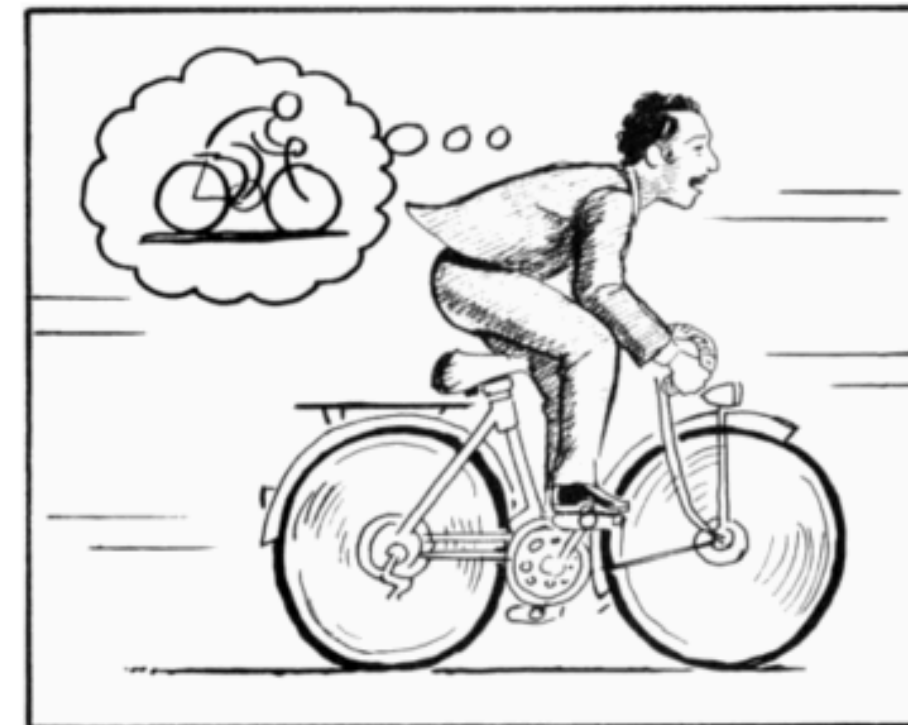


Figure 1. A World Model, from Scott McCloud's *Understanding Comics*. (McCloud, 1993; E, 2012)

current motor actions (Keller et al., 2012; Leinweber et al.,

World Models(2018)

At each time step, our agent receives an **observation** from the environment.

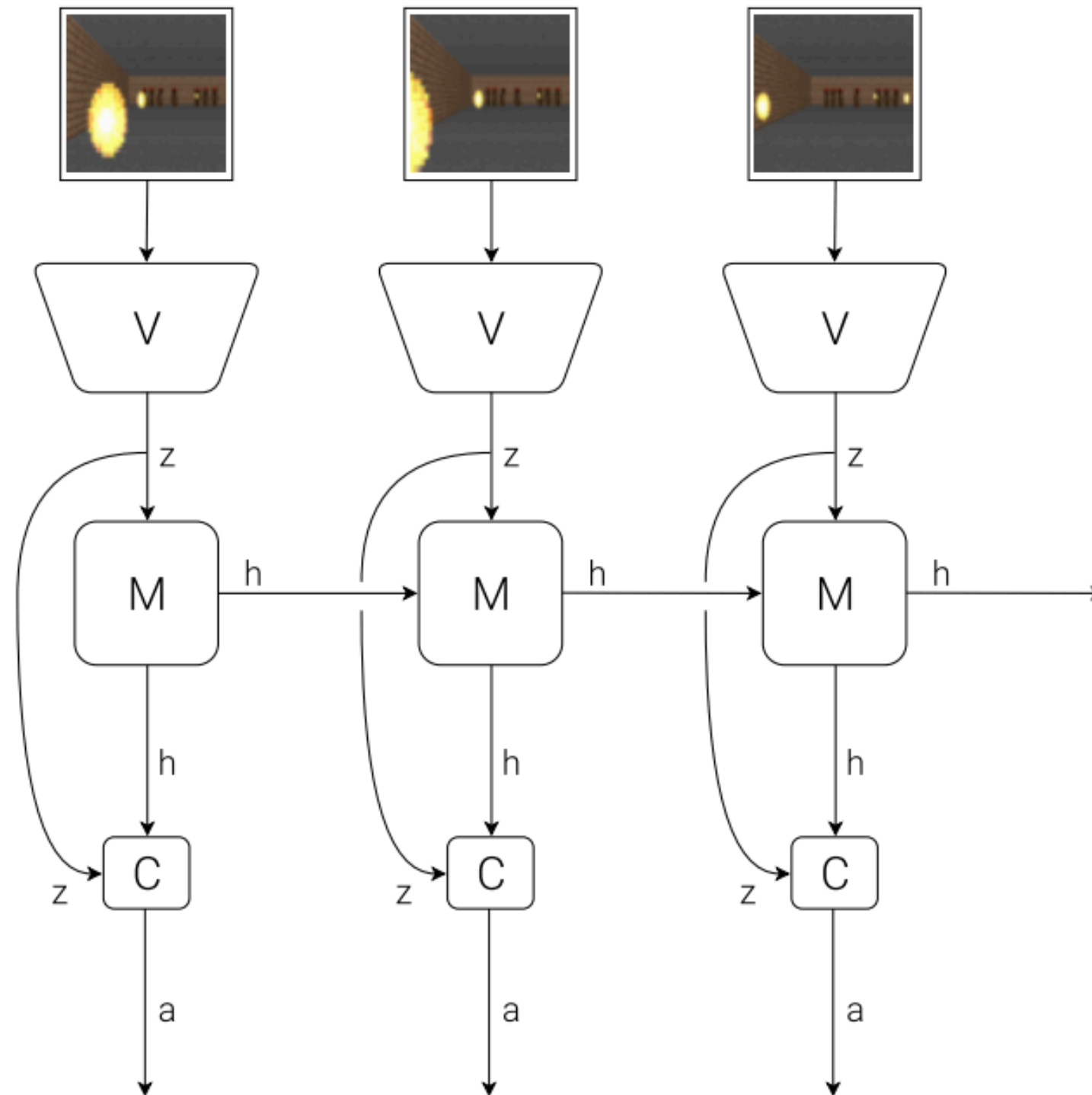
World Model

The **Vision Model (V)** encodes the high-dimensional observation into a low-dimensional latent vector.

The **Memory RNN (M)** integrates the historical codes to create a representation that can predict future states.

A small **Controller (C)** uses the representations from both **V** and **M** to select good actions.

The agent performs **actions** that go back and affect the environment.



World Models(2018)

Vision Model(V) – ConvVAE

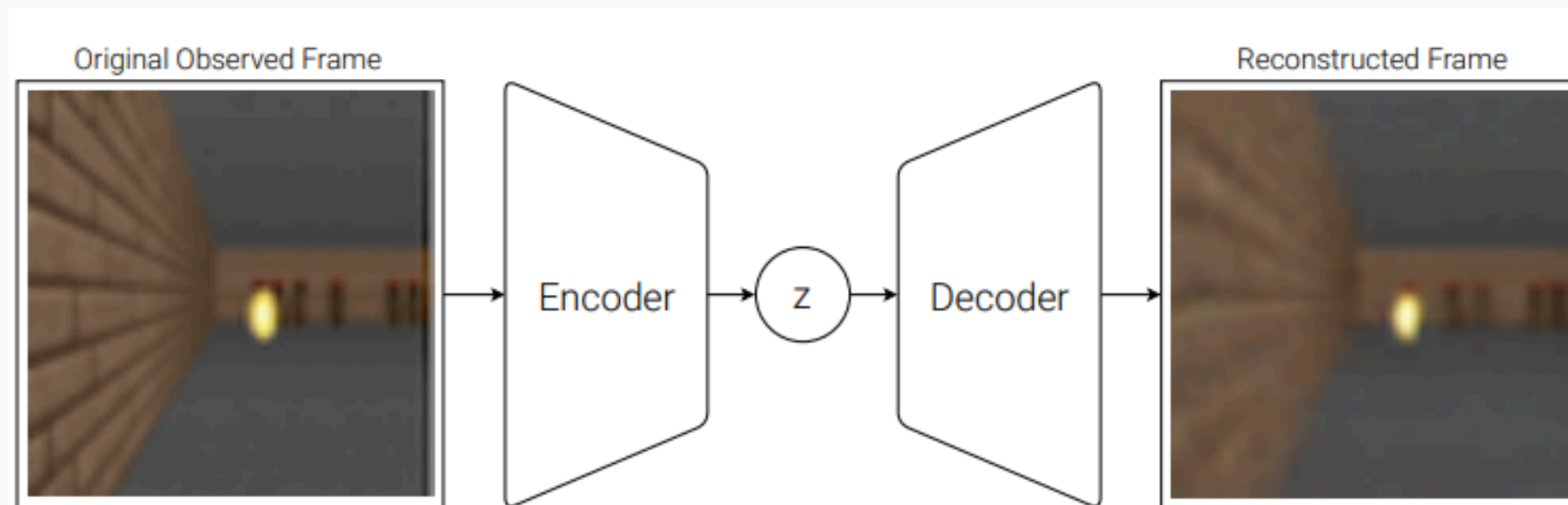
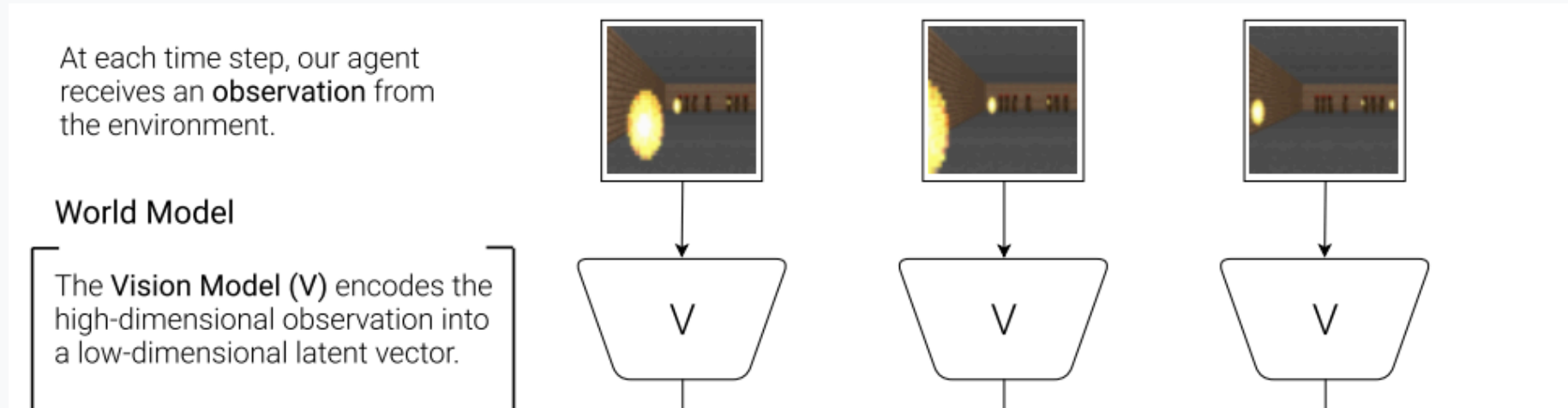


Figure 5. Flow diagram of a Variational Autoencoder (VAE).

A.1. Variational Autoencoder

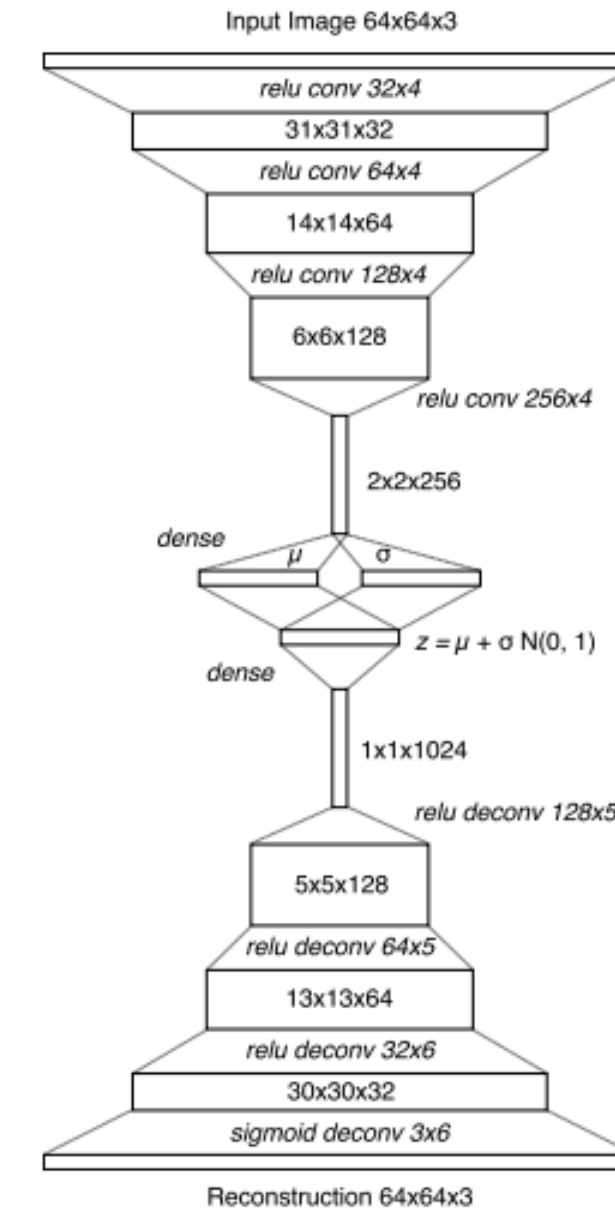


Figure 22. Description of tensor shapes at each layer of ConvVAE.

World Models(2018)

Memory (M) – MDN-RNN

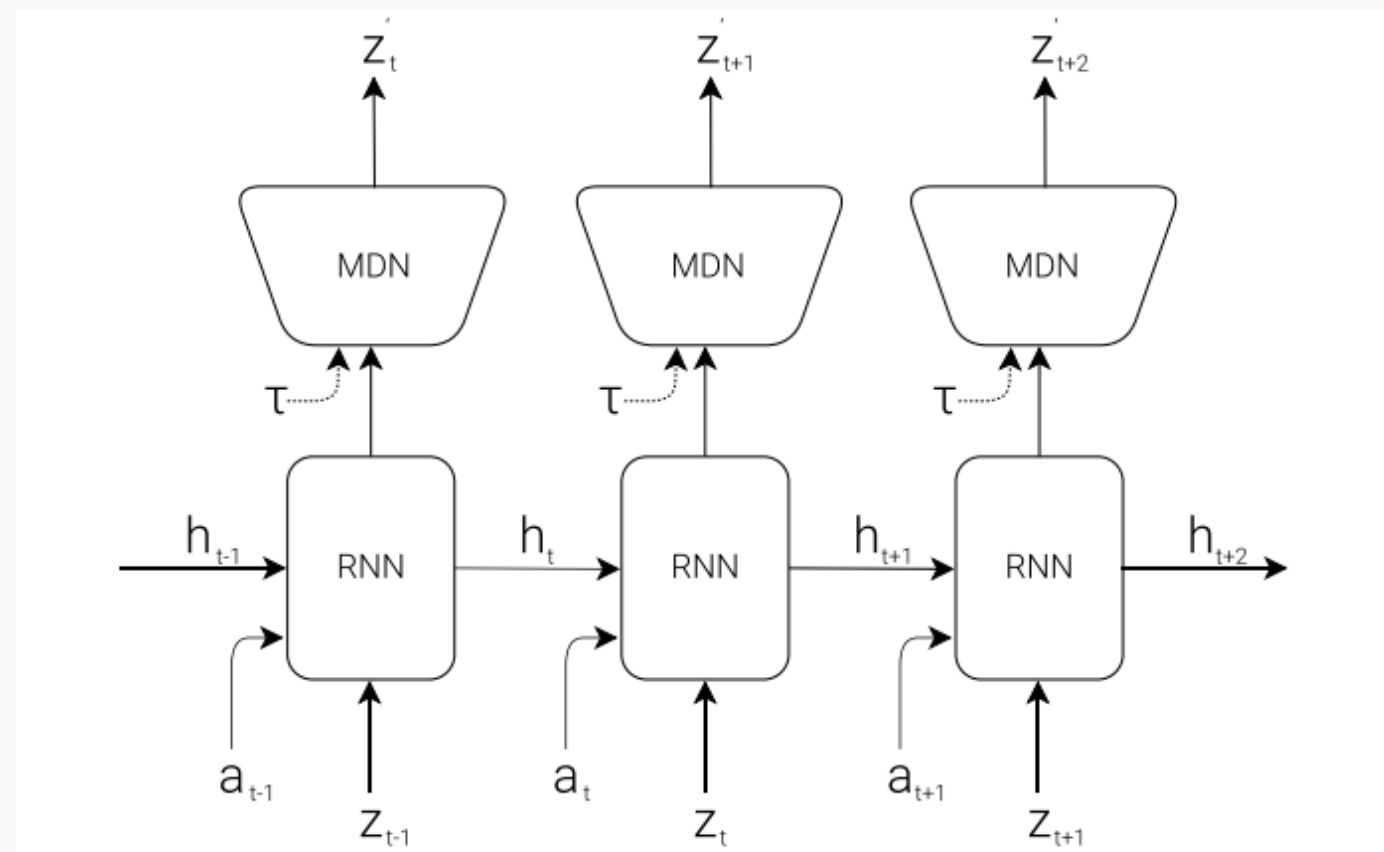
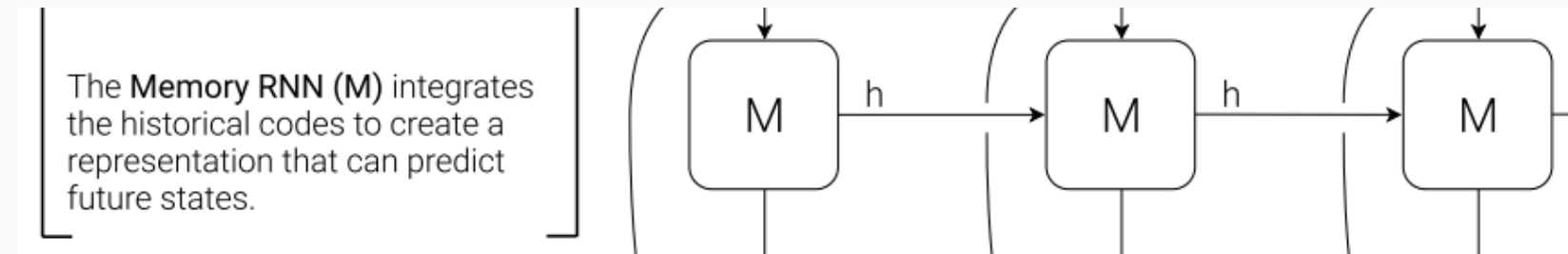


Figure 6. RNN with a Mixture Density Network output layer. The MDN outputs the parameters of a mixture of Gaussian distribution used to sample a prediction of the next latent vector z .

Controller(C)

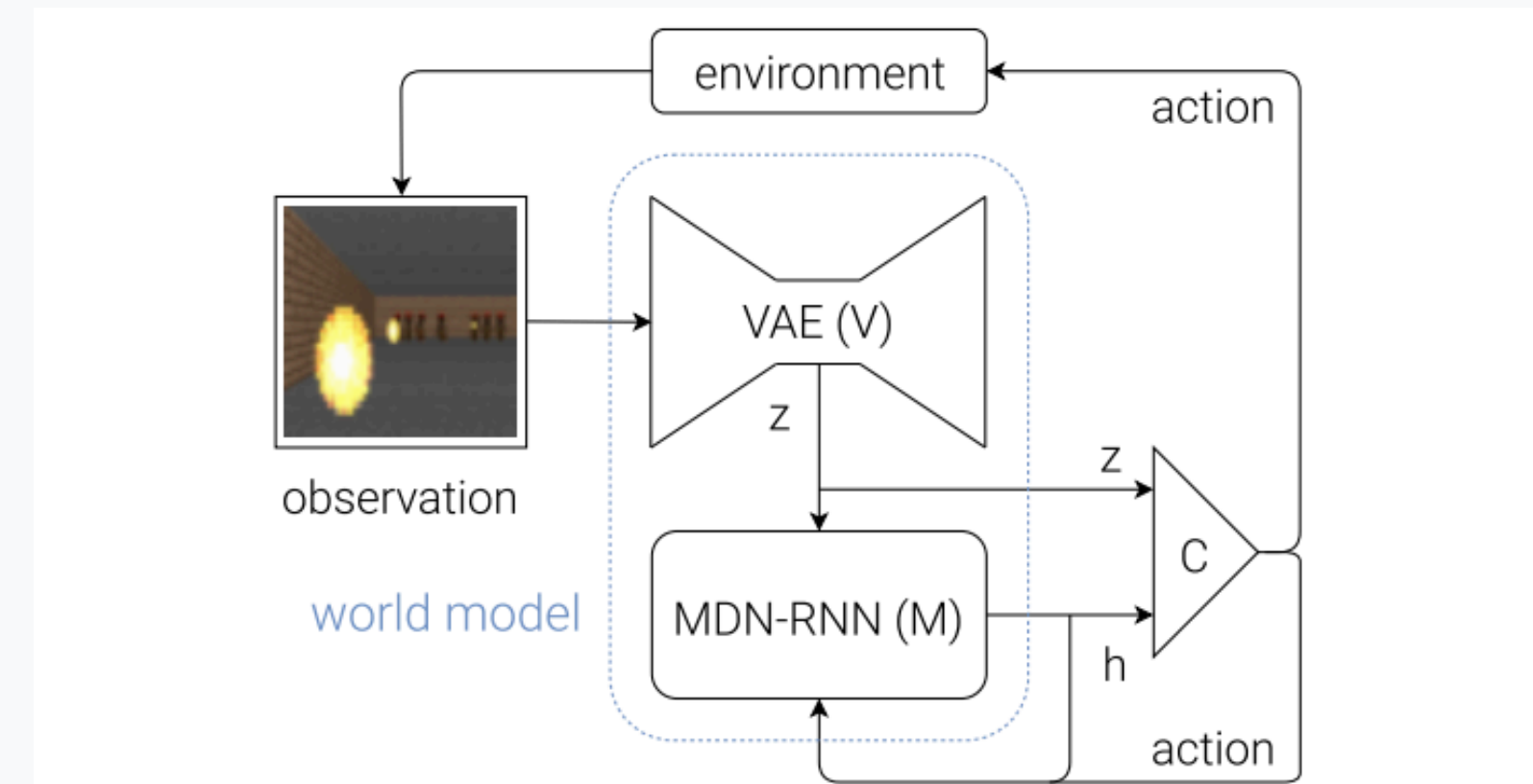


Figure 8. Flow diagram of our Agent model. The raw observation is first processed by V at each time step t to produce z_t . The input into C is this latent vector z_t concatenated with M 's hidden state h_t at each time step. C will then output an action vector a_t for motor control, and will affect the environment. M will then take the current z_t and action a_t as an input to update its own hidden state to produce h_{t+1} to be used at time $t + 1$.

Result - interactive version

한계: **self-supervised learning**은 추구하지만 정작 환경에 따른 행동 라벨은 필수적.
모델 한계로 인해 한 모델이 하나의 게임만 학습함.

Genie

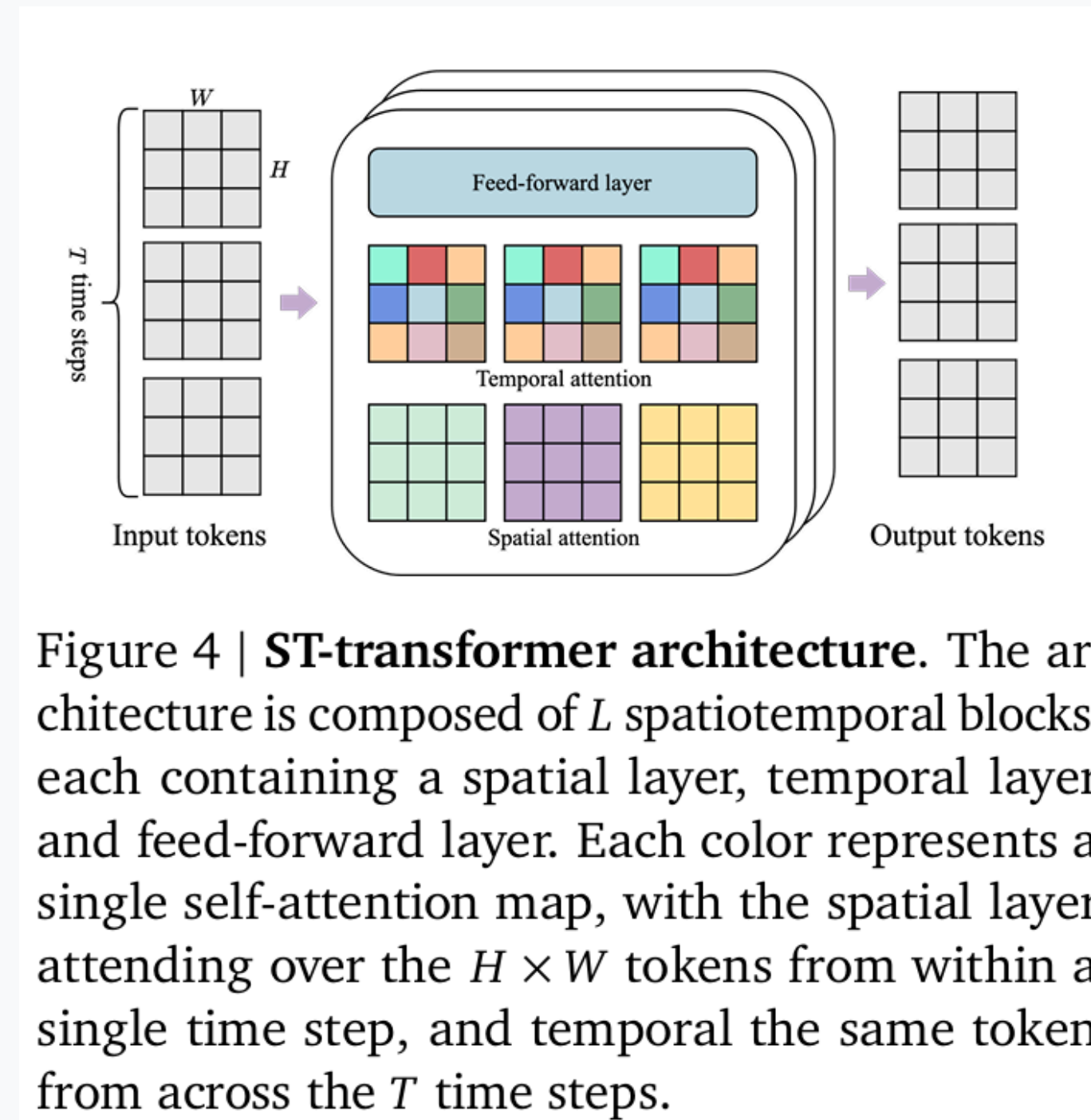
Genie: Generative Interactive Enviroments(2024)



Figure 1 | **A whole new world:** Genie is capable of converting a variety of different prompts into interactive, playable environments that can be easily created, stepped into, and explored. This is made possible via a latent action interface, learned fully unsupervised from Internet videos. On the right we see a few generated steps for taking two latent actions. See more examples on our [website](#).

Genie: Generative Interactive Enviroments(2024)

공통 구조 - ST-transformer



Genie: Generative Interactive Enviroments(2024)

Vision(V) – VQ-VAE

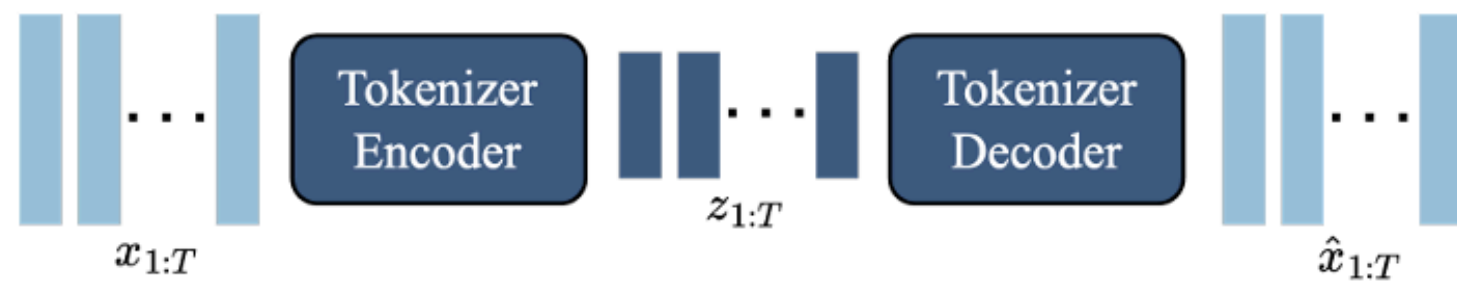


Figure 6 | **Video tokenizer:** a VQ-VAE with ST-transformer.

Memory – Dynamics Model(MaskGIT)

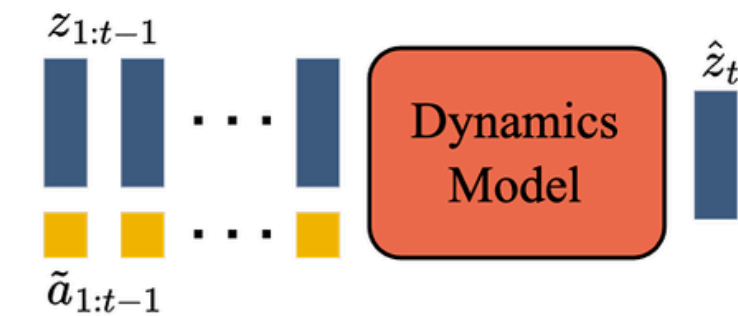


Figure 7 | **Dynamics model:** takes in video tokens and action embeddings, and predicts future masked video tokens.

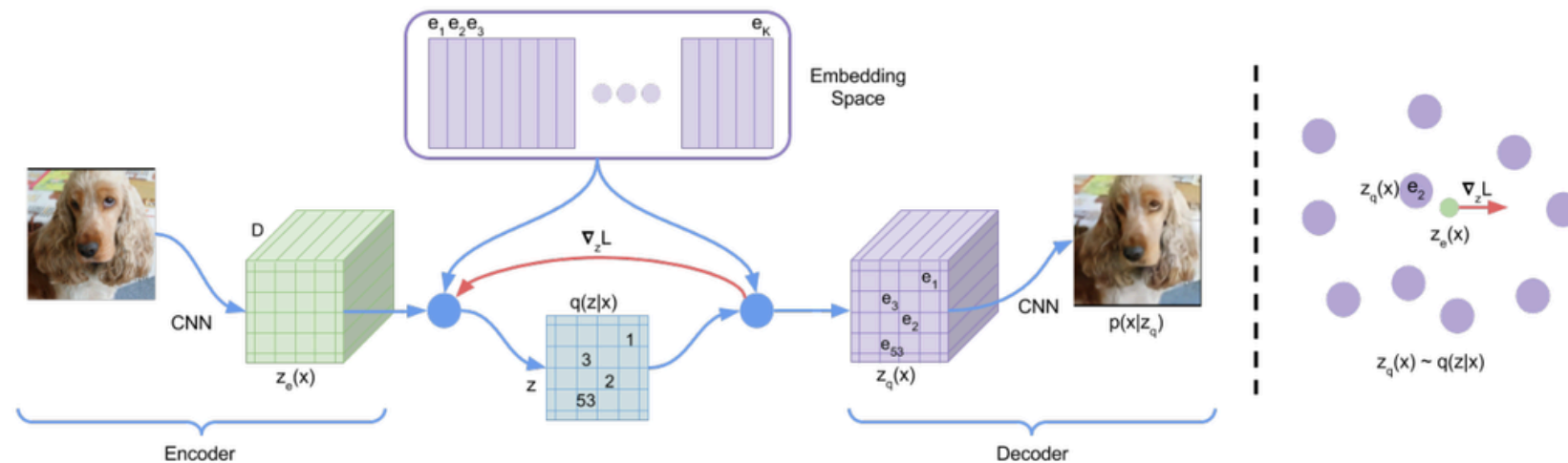


Figure 1: Left: A figure describing the VQ-VAE. Right: Visualisation of the embedding space. The output of the encoder $z(x)$ is mapped to the nearest point e_2 . The gradient $\nabla_z L$ (in red) will push the encoder to change its output, which could alter the configuration in the next forward pass.

Genie: Generative Interactive Enviroments(2024)

LAM(Latent Action Model) & Inference

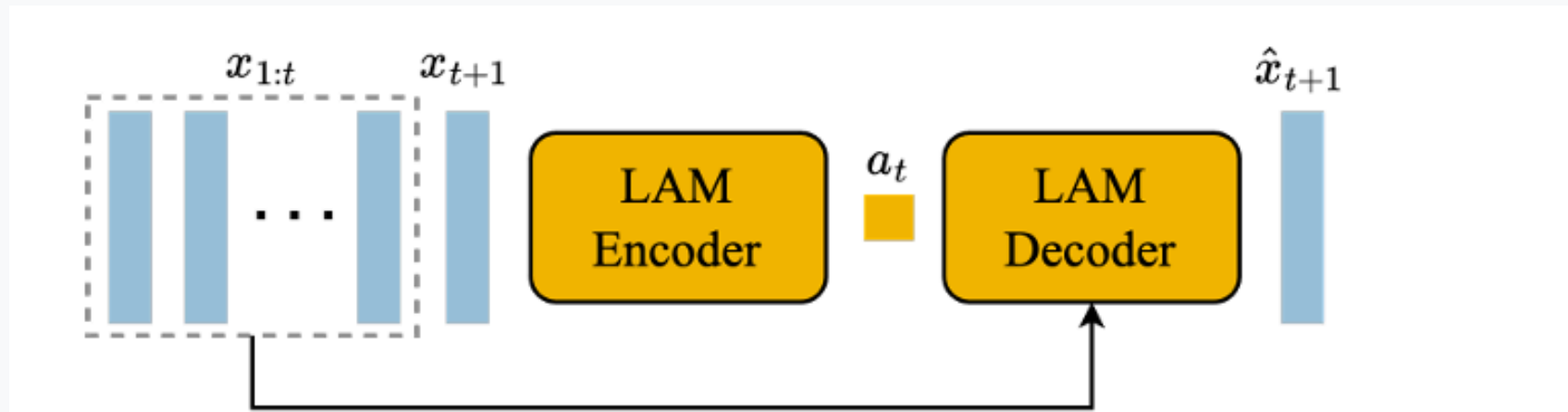


Figure 5 | **Latent action model:** learns actions a_t unsupervised from unlabelled video frames.

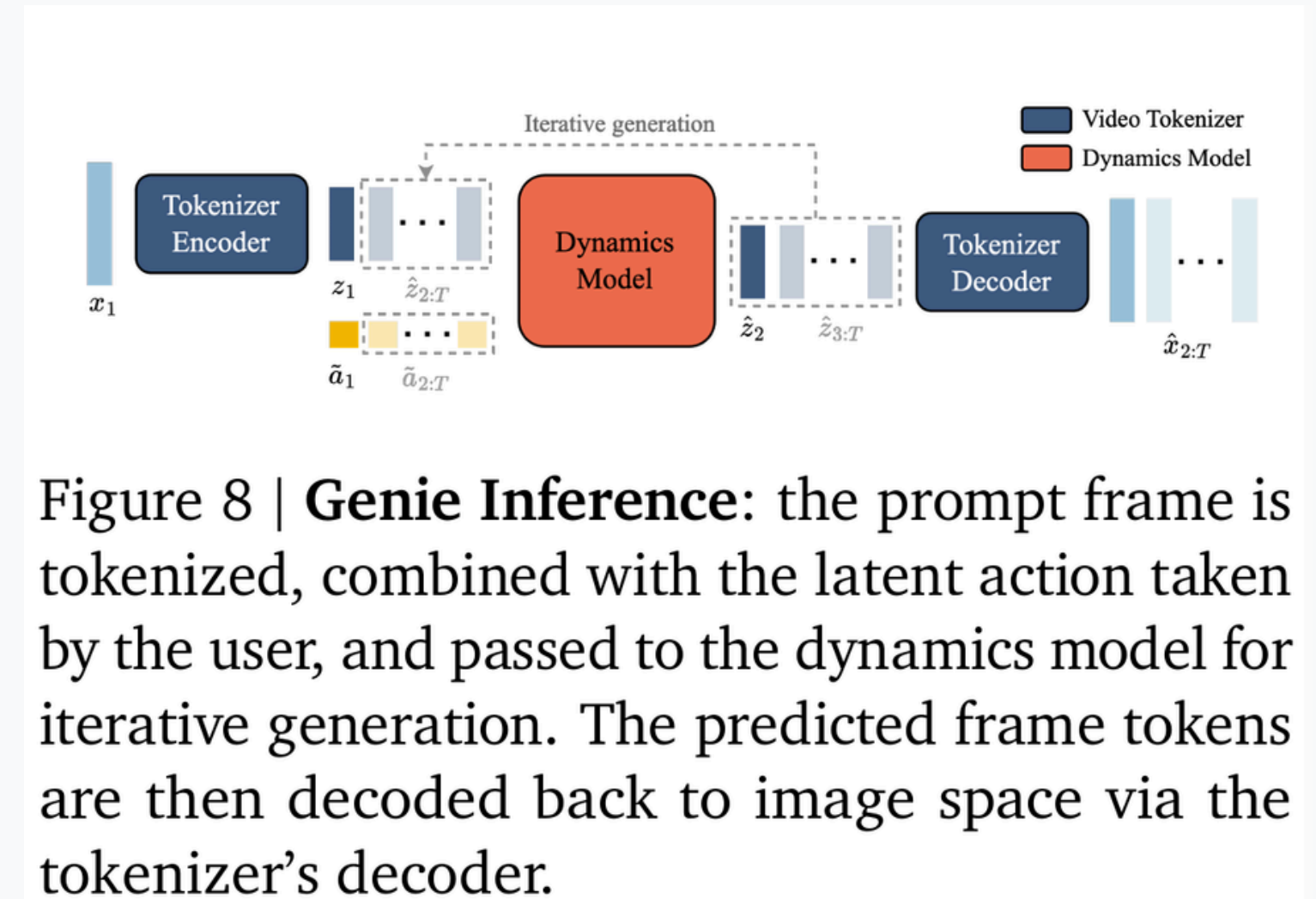


Figure 8 | **Genie Inference:** the prompt frame is tokenized, combined with the latent action taken by the user, and passed to the dynamics model for iterative generation. The predicted frame tokens are then decoded back to image space via the tokenizer's decoder.

Result - interactive version

다른 예시

결과 및 한계: LAM을 통해 실제 행동 라벨 없이 학습.

chain 구조가 그렇듯 오차가 쌓이면 붕괴함.

생성하는 데에 용량을 너무 낭비하는 경향이 있음.

JEPA

JEPA: Joint Embedding Predictive Architecture



안르쿤(Yann LeCun)

v3 [cs.CV] 13 Apr 2023

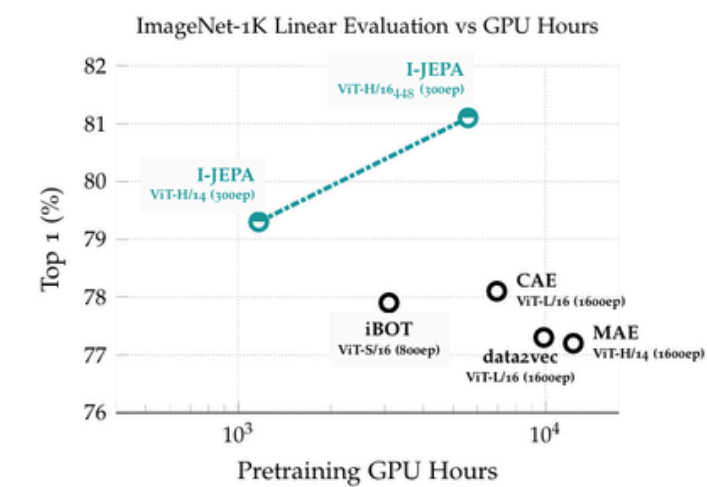
Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture

Mahmoud Assran^{1,2,3*} Quentin Duval¹ Ishan Misra¹ Piotr Bojanowski¹
Pascal Vincent¹ Michael Rabbat^{1,3} Yann LeCun^{1,4} Nicolas Ballas¹

¹Meta AI (FAIR) ²McGill University ³Mila, Quebec AI Institute ⁴New York University

Abstract

This paper demonstrates an approach for learning highly semantic image representations without relying on hand-crafted data-augmentations. We introduce the Image-based Joint-Embedding Predictive Architecture (I-JEPA), a non-generative approach for self-supervised learning from images. The idea behind I-JEPA is simple: from a single context block, predict the representations of various target blocks in the same image. A core design choice to guide I-JEPA towards producing semantic representations is the masking strategy; specifically, it is crucial to (a) sample target blocks with sufficiently large scale (semantic), and to (b) use a sufficiently informative (spatially distributed) context



JEPA: Joint Embedding Predictive Architecture

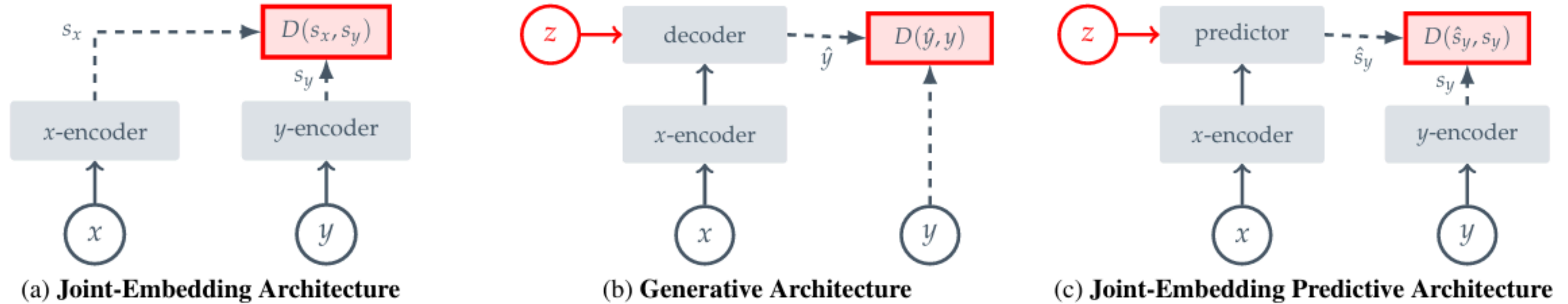


Figure 2. Common architectures for self-supervised learning, in which the system learns to capture the relationships between its inputs. The objective is to assign a high energy (large scalar value) to incompatible inputs, and to assign a low energy (low scalar value) to compatible inputs. **(a)** Joint-Embedding Architectures learn to output similar embeddings for compatible inputs x, y and dissimilar embeddings for incompatible inputs. **(b)** Generative Architectures learn to directly reconstruct a signal y from a compatible signal x , using a decoder network that is conditioned on additional (possibly latent) variables z to facilitate reconstruction. **(c)** Joint-Embedding Predictive Architectures learn to predict the embeddings of a signal y from a compatible signal x , using a predictor network that is conditioned on additional (possibly latent) variables z to facilitate prediction.

JEPA: Joint Embedding Predictive Architecture

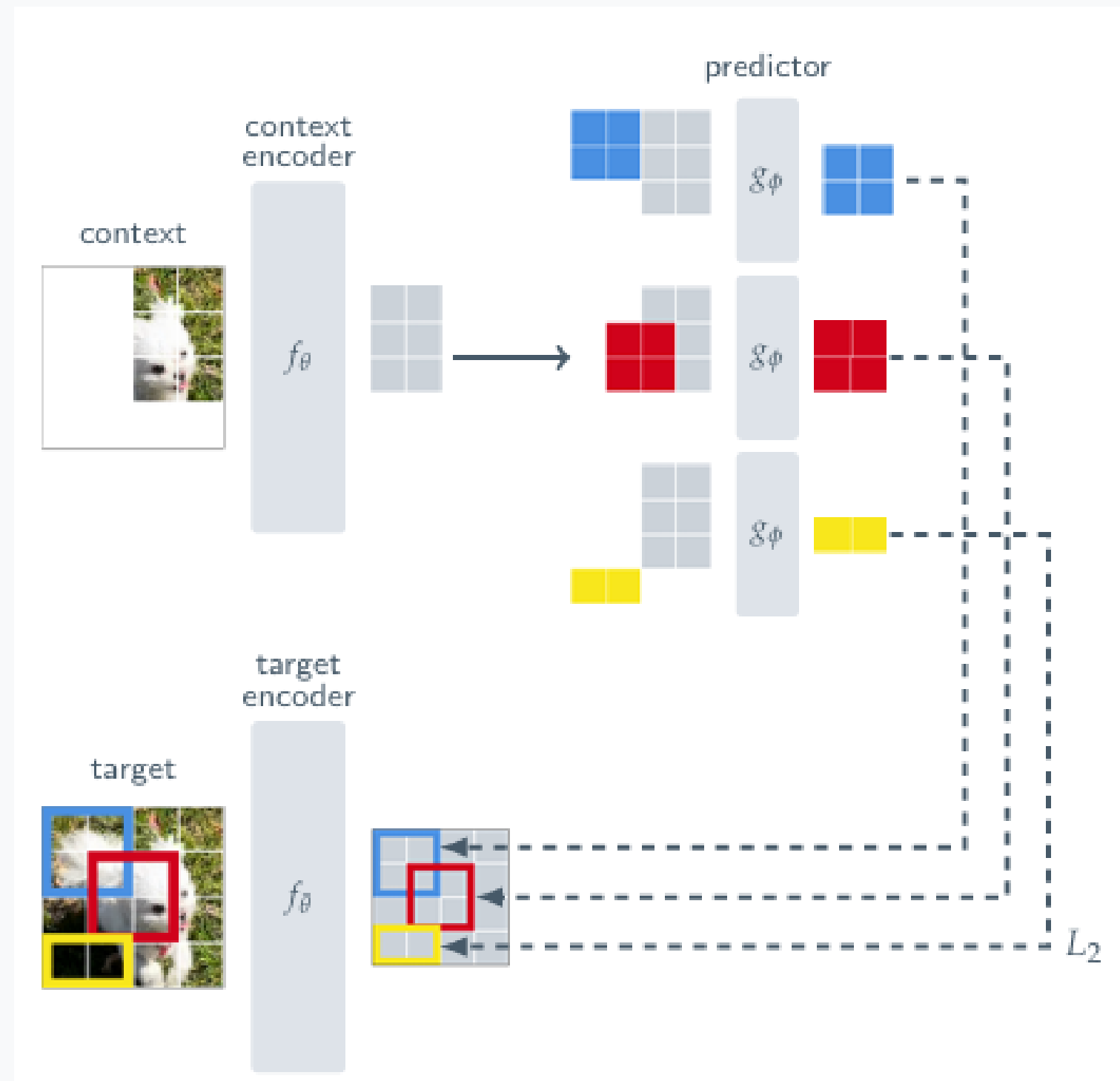


Figure 4. Examples of our context and target-masking strategy. Given an image, we randomly sample 4 target blocks with scale in the range $(0.15, 0.2)$ and aspect ratio in the range $(0.75, 1.5)$. Next, we randomly sample a context block with scale in the range $(0.85, 1.0)$ and remove any overlapping target blocks. Under this strategy, the target-blocks are relatively semantic, and the context-block is informative, yet sparse (efficient to process).

RL 기반 월드모델

DWARKESH PATEL



분명한 '올바른 행동'이 존재합니다

[한글더빙] "ChatGPT는 지능이 아닙니다" 강화학습 창시자·튜링상 수상자 리처드 서튼

사이언스 아담 Science Adam
구독자 6.12만명

가입

1.6천

공유

질문하기

저장

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감사합니다